

Step by Step ASM/ Grid Installation in Oracle Standalone

```
[root@dg11 ~]# yum install -y oracle-database-preinstall-19c
Loaded plugins: langpacks, ulninfo
ol7_UEKR6
ol7_latest
(1/4): ol7_latest/x86_64/updateinfo
(2/4): ol7_UEKR6/x86_64/updateinfo
(3/4): ol7_latest/x86_64/primary_db
(4/4): ol7_UEKR6/x86_64/primary_db
Package oracle-database-preinstall-19c-1.0-3.el7.x86_64 already installed and latest version
Nothing to do
[root@dg11 ~]# █

[root@dg11 ~]# yum -y install oracleasm*
Loaded plugins: langpacks, ulninfo
Package oracleasm-support-2.1.11-2.el7.x86_64 already installed and latest version
Nothing to do
[root@dg11 ~]# yum -y install kmod-oracleasm*
Loaded plugins: langpacks, ulninfo
Package kmod-oracleasm-2.0.8-28.0.1.el7.x86_64 already installed and latest version
Nothing to do
```

Add below OS groups

```
[root@dg11 ~]# groupadd -g 54327 asmdba
[root@dg11 ~]# groupadd -g 54328 asmoper
[root@dg11 ~]# groupadd -g 54329 asmadmin
```

Add asmdba as secondary group to Oracle user

```
[root@dg11 ~]# usermod -a -G asmdba oracle
```

Create Grid User

```
1 [root@dg11 /]# useradd -u 54331 -g oinstall -G dba,asmdba,asmoper,asmadmin,racdba g
```

Change the password for Oracle and Grid user

```
[root@dg11 /]# passwd oracle
Changing password for user oracle.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@dg11 /]# passwd grid
Changing password for user grid.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
```

Create the Directories for Oracle Database installation

```
[root@dg11 /]# mkdir -p /u01/app/oracle
[root@dg11 /]#
[root@dg11 /]# mkdir -p /u01/app/oracle/product/19/db
[root@dg11 /]#
[root@dg11 /]# chown -R oracle:oinstall /u01
```

Create the Directories for Oracle Grid installation

```
[root@dg11 /]# mkdir -p /u01/app/grid/product/19/grid
[root@dg11 /]#
[root@dg11 /]# chown -R grid:oinstall /u01/app/grid
[root@dg11 /]#
[root@dg11 /]# chmod -R 775 /u01
```

Step2. Configure Oracle ASM

```
[root@dg11 /]# oracleasm configure -i
Configuring the Oracle ASM library driver.
```

This will configure the on-boot properties of the Oracle ASM library driver. The following questions will determine whether the driver is loaded on boot and what permissions it will have. The current values will be shown in brackets ('[]'). Hitting <ENTER> without typing an answer will keep that current value. Ctrl-C will abort.

```
Default user to own the driver interface [grid]: grid
Default group to own the driver interface [oinstall]: oinstall
Start Oracle ASM library driver on boot (y/n) [y]: y
Scan for Oracle ASM disks on boot (y/n) [y]: y
Writing Oracle ASM library driver configuration: done
```

Load / initiate Oracle ASM

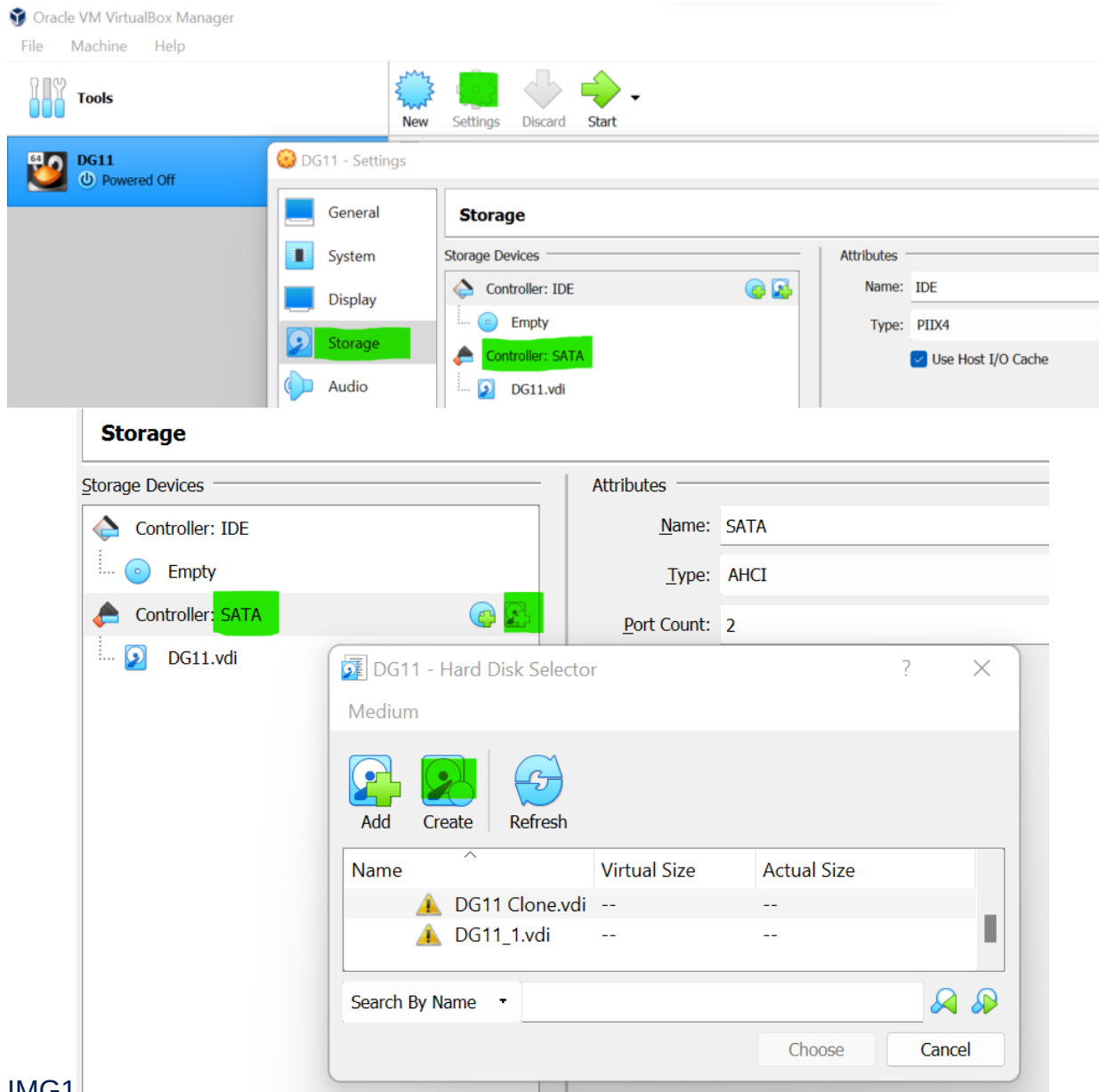
```
[root@dg11 /]# sync
[root@dg11 /]# oracleasm init [root@dg11 /]# shutdown -r now
```

init 0

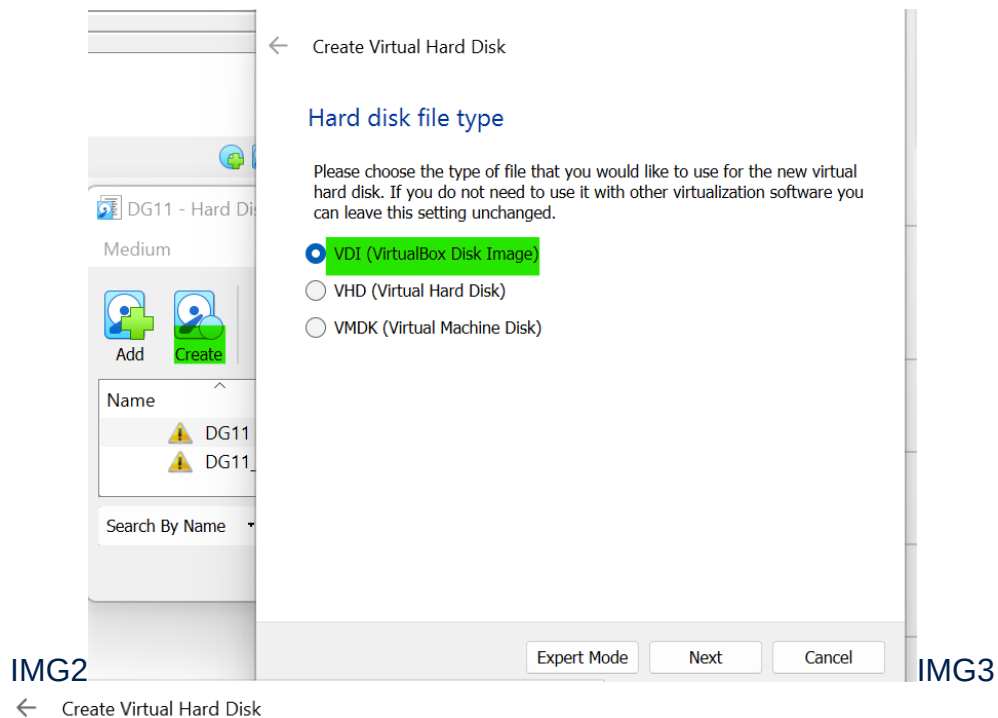
Add 25 GB Hard Disk

Add 25 GB HDD to virtual machine which we will use to create 3 partitions of 5,10,10 GB each for our ASM disks.

Open Virtual machine >> Settings >> Storage >> Controller: SATA >> Create new disk



IMG1



Storage on physical hard disk

Please choose whether the new virtual hard disk file should grow as it is used (dynamically allocated) or if it should be created at its maximum size (fixed size).

A **dynamically allocated** hard disk file will only use space on your physical hard disk as it fills up (up to a maximum **fixed size**), although it will not shrink again automatically when space on it is freed.

A **fixed size** hard disk file may take longer to create on some systems but is often faster to use.

☐ Dynamically allocated

☒ Fixed size



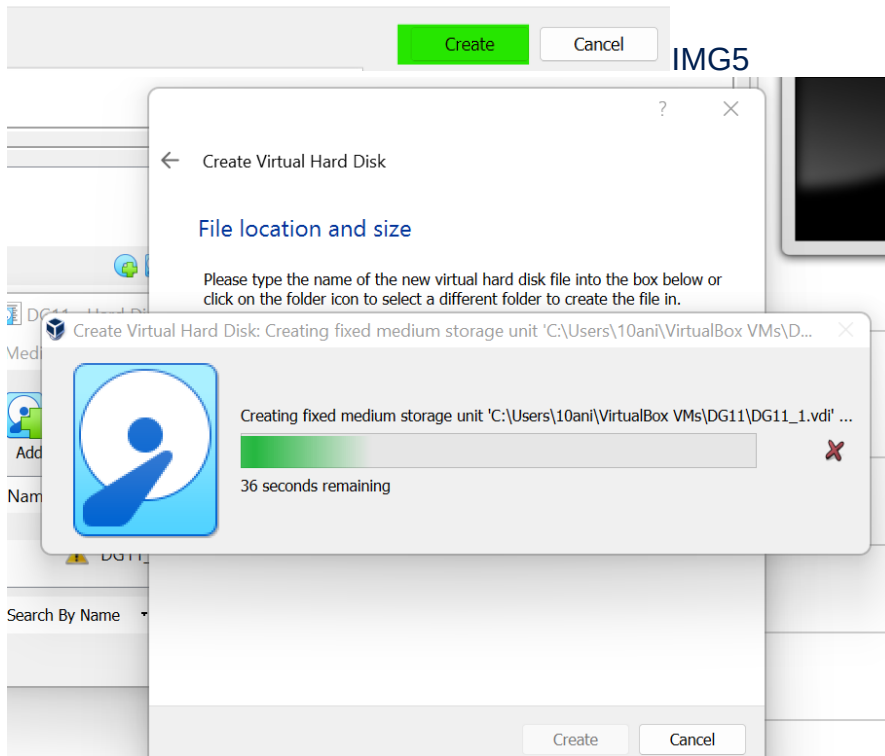


File location and size

Please type the name of the new virtual hard disk file into the box below or click on the folder icon to select a different folder to create the file in.

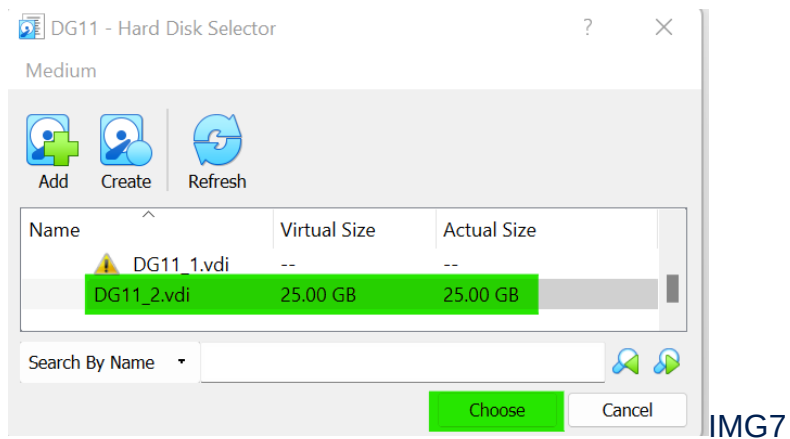
C:\Users\10ani\VirtualBox VMs\DG11\DG11_1.vdi

Select the size of the virtual hard disk in megabytes. This size is the limit on the amount of file data that a virtual machine will be able to store on the hard disk.



IMG5

IMG6



Click on OK & Start the VM Machine.

Start virtual machine and login as root user to format the newly added disk

```
[root@dg11 ~]# fdisk -l
```

```
Disk /dev/sda: 103.1 GB, 103079215104 bytes, 201326592 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk label type: dos
Disk identifier: 0x000eba3a
```

Device	Boot	Start	End	Blocks	Id	System
/dev/sda1	*	2048	2099199	1048576	83	Linux
/dev/sda2		2099200	201326591	99613696	8e	Linux LVM

```
Disk /dev/sdb: 26.8 GB, 26843545600 bytes, 52428800 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disk /dev/mapper/ol-root: 53.7 GB, 53687091200 bytes, 104857600 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disk /dev/mapper/ol-swap: 3221 MB, 3221225472 bytes, 6291456 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disk /dev/mapper/ol-home: 45.1 GB, 45088768000 bytes, 88064000 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
[root@dg11 ~]#
```

IMG8

```
[root@dg11 ~]# fdisk /dev/sdb
Welcome to fdisk (util-linux 2.23.2).
```

```
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.
```

```
Device does not contain a recognized partition table
Building a new DOS disklabel with disk identifier 0xb333dd4d.
```

```
Command (m for help): n
```

```
Partition type:
```

```
  p   primary (0 primary, 0 extended, 4 free)
  e   extended
```

```
Select (default p): p
```

```
Partition number (1-4, default 1): 1
```

```
First sector (2048-52428799, default 2048):
```

```
Using default value 2048
```

```
Last sector, +sectors or +size{K,M,G} (2048-52428799, default 52428799): +5G
```

```
Partition 1 of type Linux and of size 5 GiB is set
```

```
Command (m for help):
```

IMG9

```

Command (m for help): n
Partition type:
   p   primary (1 primary, 0 extended, 3 free)
   e   extended
Select (default p):
Using default response p
Partition number (2-4, default 2):
First sector (10487808-52428799, default 10487808):
Using default value 10487808
Last sector, +sectors or +size{K,M,G} (10487808-52428799, default 52428799): +10G
Partition 2 of type Linux and of size 10 GiB is set

Command (m for help): N
Partition type:
   p   primary (2 primary, 0 extended, 2 free)
   e   extended
Select (default p):
Using default response p
Partition number (3,4, default 3):
First sector (31459328-52428799, default 31459328):
Using default value 31459328
Last sector, +sectors or +size{K,M,G} (31459328-52428799, default 52428799): +10G
Value out of range.
Last sector, +sectors or +size{K,M,G} (31459328-52428799, default 52428799):
Using default value 52428799
Partition 3 of type Linux and of size 10 GiB is set

Command (m for help): w
The partition table has been altered!

Calling ioctl() to re-read partition table.
Syncing disks.
[root@dg11 ~]#

```

IMG10

Step3. Create separate ASM Disk for each partition

```

[root@dg11 ~]# oracleasm createdisk GRID_DG /dev/sdb1
Writing disk header: done
Instantiating disk: done
[root@dg11 ~]# oracleasm createdisk PROD_DG1 /dev/sdb2
Writing disk header: done
Instantiating disk: done
[root@dg11 ~]# oracleasm createdisk PROD_DG_FLASH /dev/sdb3
Writing disk header: done
Instantiating disk: done
[root@dg11 ~]#

```

To check asm disks

```

[root@dg11 ~]# oracleasm listdisks
GRID_DG
PROD_DG1
PROD_DG_FLASH
[root@dg11 ~]# ls -lrt /dev/oracleasm/disks
total 0
brw-rw----. 1 grid oinstall 8, 17 Nov 19 16:06 GRID_DG
brw-rw----. 1 grid oinstall 8, 18 Nov 19 16:06 PROD_DG1
brw-rw----. 1 grid oinstall 8, 19 Nov 19 16:06 PROD_DG_FLASH
[root@dg11 ~]#

```

Step4. Install Grid Software

```

[root@dg11 ~]# su - grid
Last login: Sat Nov 19 16:10:40 IST 2022
[grid@dg11 ~]$ whoami
grid
[grid@dg11 ~]$ ls -ltrh .bash_profile
-rw-r--r--. 1 grid oinstall 648 Sep 24 01:04 .bash_profile
[grid@dg11 ~]$
[grid@dg11 ~]$ cp .bash_profile .bash_profile_bkp

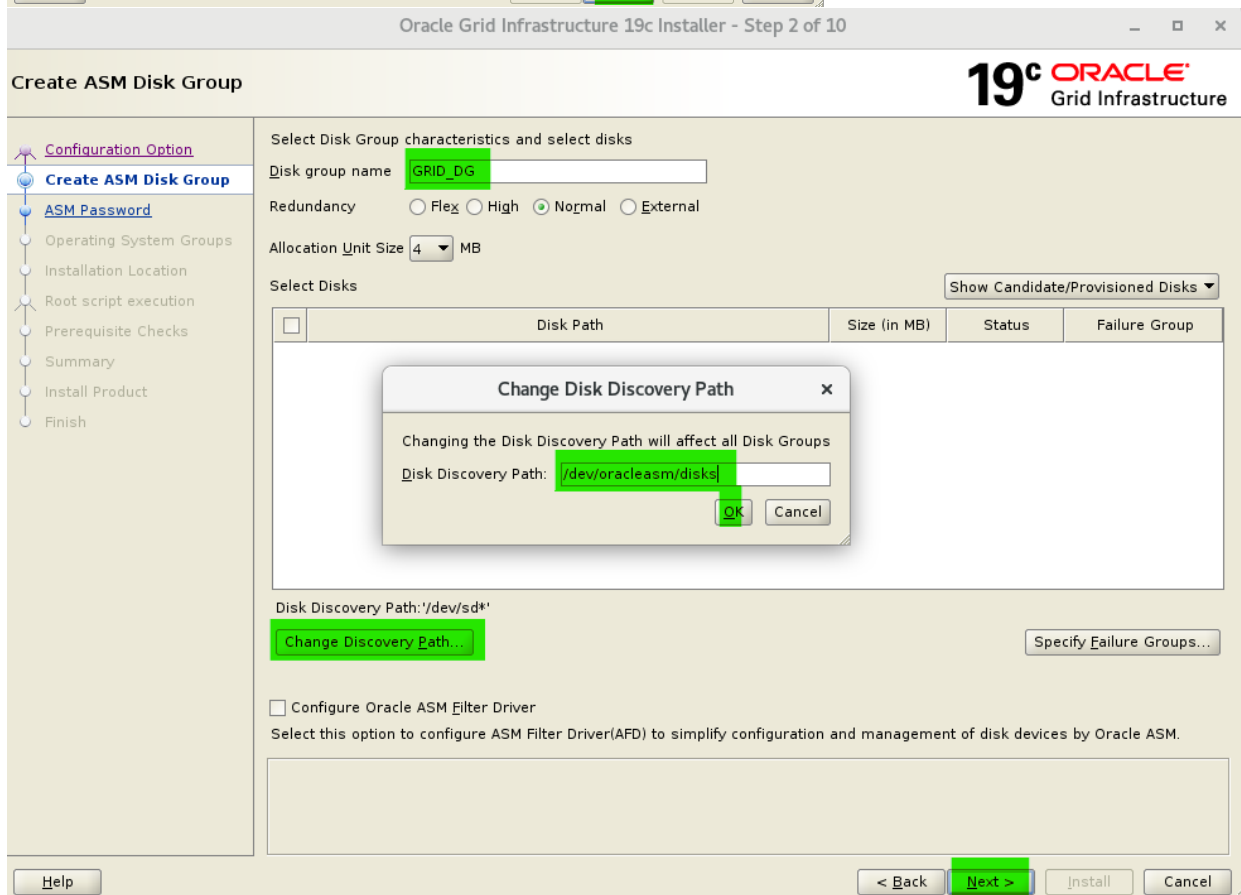
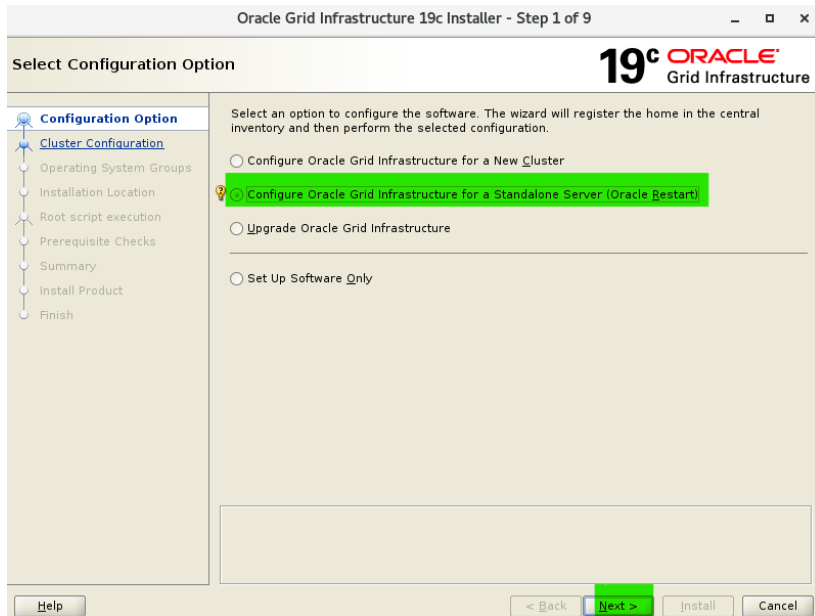
```


Put the content under vi text editor in bash profile & save it.

```
1 [grid@dg11 ~]$ vi .bash_profile
1
2 if [ -f ~/.bashrc ]; then
3   . ~/.bashrc
4 fi
5 ORACLE_SID=+ASM; export ORACLE_SID
6 ORACLE_BASE=/u01/app/grid; export ORACLE_BASE
7 ORACLE_HOME=/u01/app/grid/product/19/grid; export ORACLE_HOME
8 ORACLE_TERM=xterm; export ORACLE_TERM
9 JAVA_HOME=/usr/bin/java; export JAVA_HOME
10 TNS_ADMIN=$ORACLE_HOME/network/admin; export TNS_ADMIN
11
12 PATH=.:${JAVA_HOME}/bin:${PATH}:${HOME}/bin:${ORACLE_HOME}/bin
13 PATH=${PATH}:/usr/bin:/bin:/usr/local/bin
14 export PATH
15
16 [grid@dg11 ~]$ . .bash_profile
17 [grid@dg11 ~]$
18 [grid@dg11 ~]$ env | grep ORA
ORACLE_SID=+ASM
ORACLE_BASE=/u01/app/grid
ORACLE_TERM=xterm
ORACLE_HOME=/u01/app/grid/product/19/grid
19 [grid@dg11 ~]$
20 [root@dg11 grid]# ls -ltrh
total 2.7G
-rwxr-x---. 1 root root 2.7G Nov 19 16:23 LINUX.X64_193000_grid_home.zip
21 [root@dg11 grid]#
22 [root@dg11 grid]# pwd
/u01/app/grid/product/19/grid
23 [root@dg11 grid]#
24 [root@dg11 grid]# whoami
root
25 [root@dg11 grid]# unzip LINUX.X64_193000_grid_home.zip
26 [root@dg11 19]# ls -ltrh
total 4.0K
drwxrwxr-x. 65 grid oinstall 4.0K Nov 19 16:26 grid
27 [root@dg11 19]# pwd
/u01/app/grid/product/19
28 [root@dg11 19]#
29 [root@dg11 19]# whoami
root
30 [root@dg11 19]#
31 [root@dg11 19]# chown -R grid:oinstall grid
32 [root@dg11 19]#
```

Restart your server to overcome graphics issues. (init 6 or shutdown -r now)

```
[grid@dg11 ~]$ cd /u01/app/grid/product/19/grid/
[grid@dg11 grid]$
[grid@dg11 grid]$ ls -ltrh gridSetup.sh
-rwxr-x---. 1 grid oinstall 3.3K Mar  8 2017 gridSetup.sh
[grid@dg11 grid]$
[grid@dg11 grid]$ ./gridSetup.sh
```



Oracle Grid Infrastructure 19c Installer - Step 2 of 10

Create ASM Disk Group

19c ORACLE®
Grid Infrastructure

Configuration Option

Create ASM Disk Group

ASM Password

Operating System Groups

Installation Location

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

Select Disk Group characteristics and select disks

Disk group nameGRID_DG

Redundancy☐ Flex ☐ High ☐ Normal ☒ External

Allocation Unit Size4 MB

Select Disks

Show Candidate/Provisioned Disks

	Disk Path	Size (in MB)	Status
☒	/dev/oracleasm/disks/GRID_DG	5120	Provisioned
☐	/dev/oracleasm/disks/PROD_DG1	10240	Provisioned
☐	/dev/oracleasm/disks/PROD_DG_FLASH	10239	Provisioned

Disk Discovery Path: '/dev/oracleasm/disks'

Change Discovery Path...

☐ Configure Oracle ASM Filter Driver

Select this option to configure ASM Filter Driver(AFD) to simplify configuration and management of disk devices by Oracle ASM.

Help

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Next >

Install

Cancel

Oracle Grid Infrastructure 19c Installer - Step 3 of 10

Specify ASM Password

19c ORACLE®
Grid Infrastructure

Configuration Option

Create ASM Disk Group

ASM Password

Operating System Groups

Installation Location

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

The new Oracle Automatic Storage Management (Oracle ASM) instance requires its own SYS user with SYSASM privileges for administration. Oracle recommends that you create a less privileged ASMSNMP user with SYSDBA privileges to monitor the ASM instance.

Specify the password for these user accounts.

☐ Use different passwords for these accounts

	Password	Confirm Password
SYS		
ASMSNMP		

☒ Use same passwords for these accounts

Specify Password: Confirm Password:

Messages:

Specify Password:[INS-30011] The password entered does not conform to the Oracle recommended standards.

Help

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Next >

Install

Cancel

Specify the password for these user accounts.

☐ Use different passwords for these accounts

Password

Confirm Password

Oracle Grid Infrastructure 19c Installer



[INS-30011] The password entered does not conform to the Oracle recommended standards.

Are you sure you want to continue ?

Yes

No

Details

Oracle Grid Infrastructure 19c Installer - Step 4 of 11

19c ORACLE

Grid Infrastructure

Specify Management Options

Configuration Option

Create ASM Disk Group

ASM Password

Management Options

Operating System Groups

Installation Location

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

You can configure to have this instance of Oracle Grid Infrastructure and Oracle Automatic Storage Management to be managed by Enterprise Manager Cloud Control. Specify the details of the Cloud Control configuration to perform the registration.

☐ Register with Enterprise Manager (EM) Cloud Control

O~~M~~S host:

O~~M~~S port:

EM Admin User Name:

EM Admin Password:

Help

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Next >

Install

Cancel

Oracle Grid Infrastructure 19c Installer - Step 5 of 11

Privileged Operating System Groups

19c ORACLE
Grid Infrastructure

Configuration Option

Create ASM Disk Group

ASM Password

Management Options

Operating System Groups

Installation Location

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

Select the name of the operating system group, that you want to use for operating system authentication to Oracle Automatic Storage Management.

Oracle ASM Administrator (OSASM) Group

asmadmin

Oracle ASM DBA (OSDBA for ASM) Group

asmdba

Oracle ASM Operator (OSOPER for ASM) Group (Optional)

asmoper

Help

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Next >

Install

Cancel

Oracle Grid Infrastructure 19c Installer - Step 6 of 11

Specify Installation Location

19c ORACLE
Grid Infrastructure

Configuration Option

Create ASM Disk Group

ASM Password

Management Options

Operating System Groups

Installation Location

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

Specify the Oracle base. The Oracle base directory for the Oracle Grid Infrastructure installation is the location where diagnostic and administrative logs, and other logs associated with Oracle ASM and Oracle Clusterware are stored. This location would also contain files pertaining to the configuration of Oracle Clusterware.

Oracle base:

/u01/app/grid

Browse...

This software directory is the Oracle Grid Infrastructure home directory.

Software location: /u01/app/grid/product/19/grid

Help

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Next >

Install

Cancel

Oracle Grid Infrastructure 19c Installer - Step 7 of 11

19c ORACLE
Grid Infrastructure

Root script execution configuration

Configuration Option

Create ASM Disk Group

ASM Password

Management Options

Operating System Groups

Installation Location

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

During the software configuration, certain operations have to be performed as "root" user. You can choose to have the installer perform these operations automatically by specifying inputs for one of the options below. The input specified will also be used by the installer to perform additional prerequisite checks.

☐ Automatically run configuration scripts

☒ Use "root" user credential

Password :

☐ Use sudo

Program path :

User name :

Password :

Help

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Next >

Install

Cancel

Oracle Grid Infrastructure 19c Installer - Step 8 of 11

19c ORACLE
Grid Infrastructure

Perform Prerequisite Checks

Configuration Option

Create ASM Disk Group

ASM Password

Management Options

Operating System Groups

Installation Location

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

Verifying that the target environment meets minimum installation and configuration requirements for products you have selected. This can take time. Please wait.

33%

Checking Group Membership: asmdba ...

Oracle Grid Infrastructure 19c Installer - Step 8 of 11

19c ORACLE
Grid Infrastructure

Perform Prerequisite Checks

Configuration Option

Create ASM Disk Group

ASM Password

Management Options

Operating System Groups

Installation Location

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

Verification Result

Some of the minimum requirements for installation are not completed. Review and fix the issues listed in the following table, and recheck the system.

Check Again

Fix & Check Again

Show Failed

☐ Ignore All

Checks	Status	Fixable
Checks		
Physical Memory	Warning	No
Package: cvuqdisk-1.0.10-1	Warning	Yes

Oracle Grid Infrastructure 19c Installer - Step 8 of 11

19c ORACLE
Grid Infrastructure

Perform Prerequisite Checks

Configuration Option

- Create ASM Disk Group
- ASM Password
- Management Options
- Operating System Groups
- Installation Location
- Root script execution
- Prerequisite Checks**
- Summary
- Install Product
- Finish

Verification Result

Some of the minimum requirements for installation are not completed. Review and fix the issues listed in the following table, and

☐ Ignore All

Status	Fixable
Warning	No
Warning	Yes

Fixup Script

Some of the prerequisites have failed on the following nodes. Installer has generated a fixup script that needs to be run as a privileged user (root) on the listed nodes.

Script: `/tmp/GridSetupActions2022-11-19_04-41-23PM/CVU_19.0.0.0.0_grid/runfixup.sh`

Nodes: dg11

To execute the fixup script:

1. Open a terminal window
2. Login as "root"
3. Run the script
4. Return to this window and click "OK" to continue

OK Cancel

This is a prerequisite condition to test whether the package "cvuqdisk-1.0.10-1" is available on the system. [\(more details\)](#)

Expected Value : cvuqdisk-1.0.10-1
Actual Value : missing

Help < Back Next > Install Cancel

```
[root@dg11 ~]# whoami
```

```
root
```

```
[root@dg11 ~]# /tmp/GridSetupActions2022-11-19_04-41-23PM/CVU_19.0.0.0.0_grid/runfixup.sh
```

```
All Fix-up operations were completed successfully.
```

```
[root@dg11 ~]#
```

Oracle Grid Infrastructure 19c Installer - Step 8 of 11

19c ORACLE
Grid Infrastructure

Perform Prerequisite Checks

Configuration Option

- Create ASM Disk Group
- ASM Password
- Management Options
- Operating System Groups
- Installation Location
- Root script execution
- Prerequisite Checks**
- Summary
- Install Product
- Finish

Verification Result

Fixup Result

Some of the minimum requirements for installation are not completed. Review and fix the issues listed in the following table, and recheck the system.

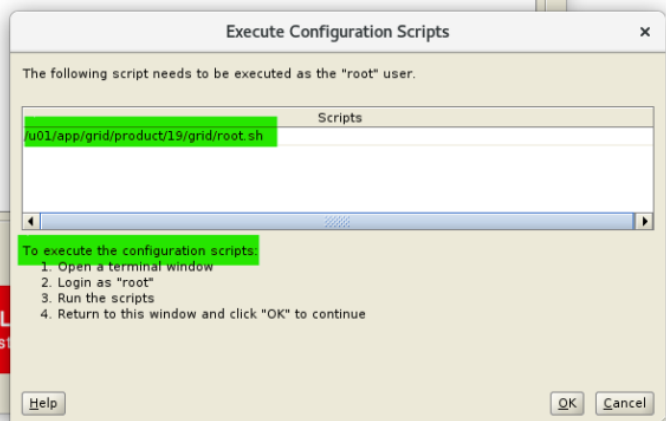
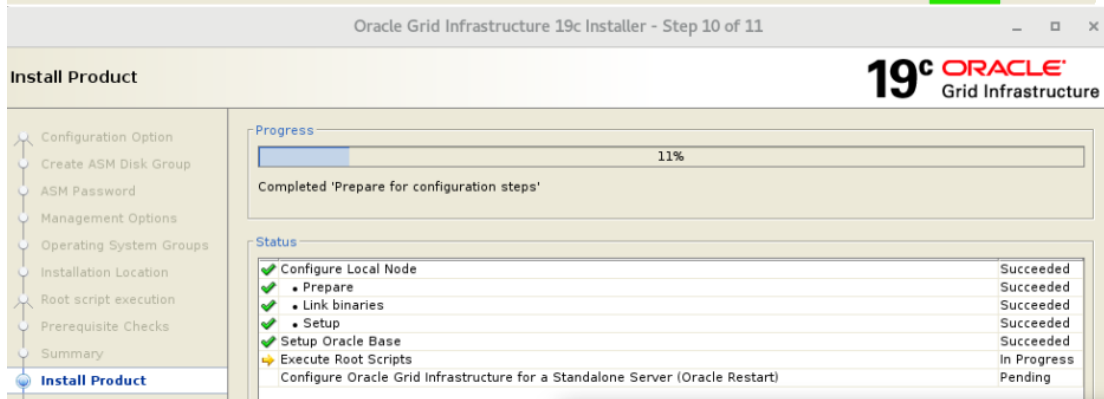
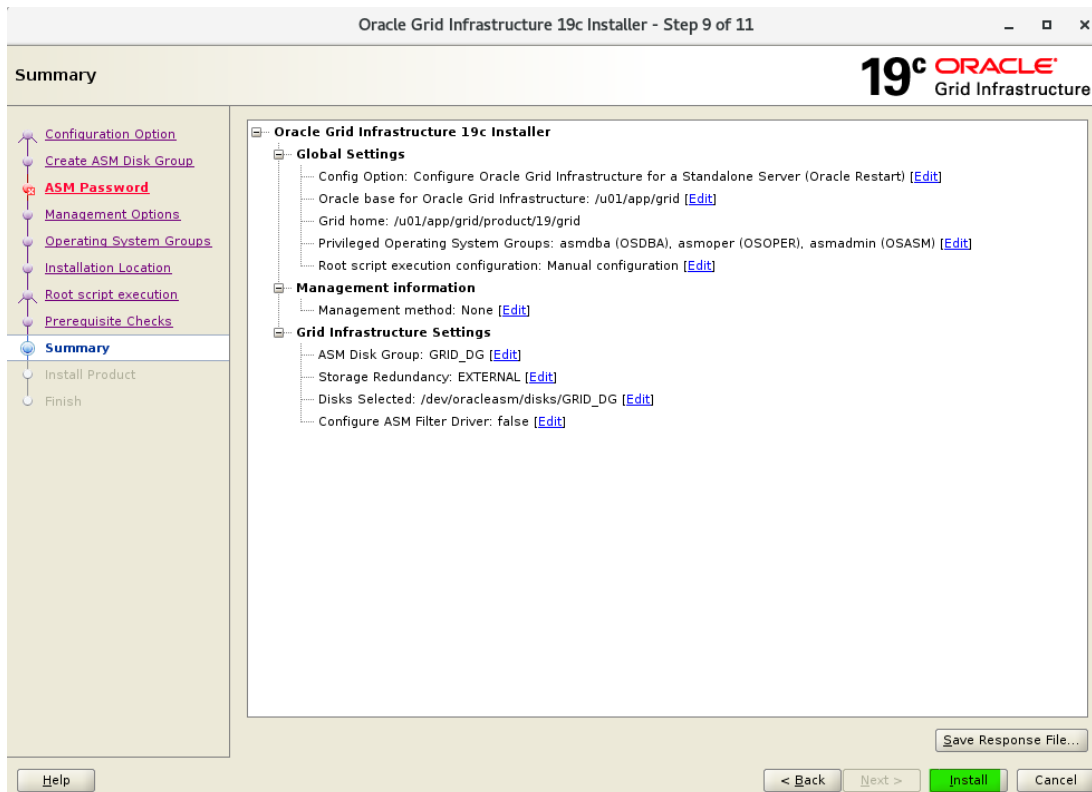
☒ Ignore All

Checks	Status	Fixable
Physical Memory	Ignored	No

This is a prerequisite condition to test whether the system has at least 8GB (8388608.0KB) of total physical memory. [\(more details\)](#)

Expected Value : 8GB (8388608.0KB)
Actual Value : 2.6401GB (2768368.0KB)

Help < Back Next > Install Cancel



```

1 [root@dg11 ~]# /u01/app/grid/product/19/grid/root.sh
2 Performing root user operation.
3
4 The following environment variables are set as:
5     ORACLE_OWNER= grid
6     ORACLE_HOME=  /u01/app/grid/product/19/grid

```

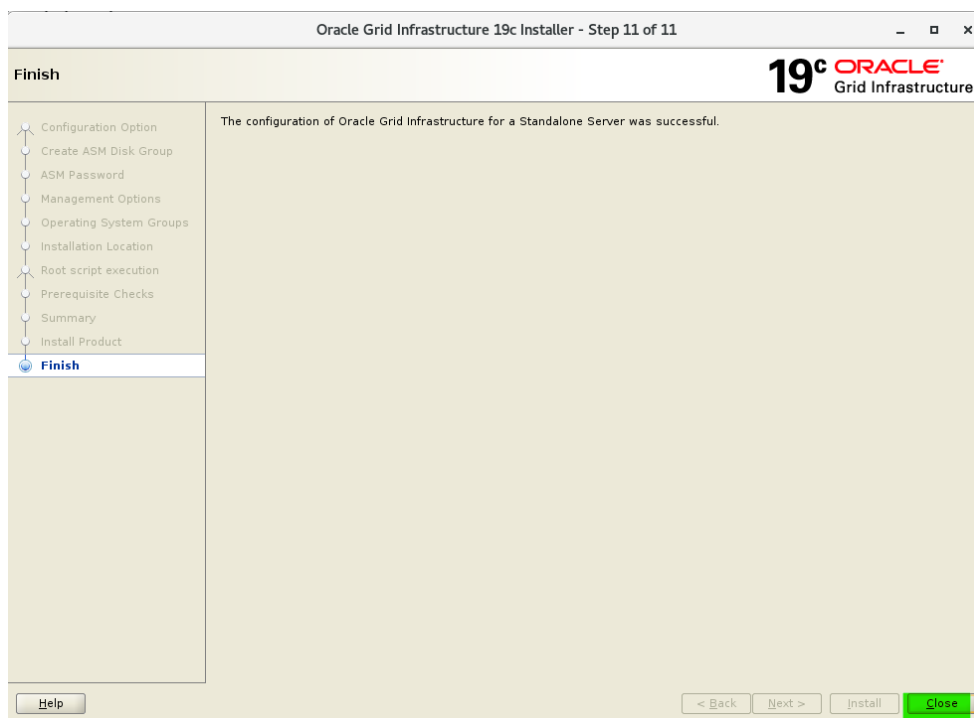


```

7   Enter the full pathname of the local bin directory: [/usr/local/bin]:
8   The contents of "dbhome" have not changed. No need to overwrite.
9   The contents of "oraenv" have not changed. No need to overwrite.
10  The contents of "coraenv" have not changed. No need to overwrite.
11
12  Entries will be added to the /etc/oratab file as needed by
13  Database Configuration Assistant when a database is created
14  Finished running generic part of root script.
15  Now product-specific root actions will be performed.
16  Using configuration parameter file: /u01/app/grid/product/19/grid/crs/install/crsconfig
17  The log of current session can be found at:
18  /u01/app/grid/crsdata/dg11/crsconfig/roothas_2022-11-19_05-00-41PM.log
19  2022/11/19 17:00:51 CLSRSC-363: User ignored prerequisites during installation
20  LOCAL ADD MODE
21  Creating OCR keys for user 'grid', privgrp 'oinstall'..
22  Operation successful.
23  LOCAL ONLY MODE
24  Successfully accumulated necessary OCR keys.
25  Creating OCR keys for user 'root', privgrp 'root'..
26  Operation successful.
27  CRS-4664: Node dg11 successfully pinned.
28  2022/11/19 17:01:05 CLSRSC-330: Adding Clusterware entries to file 'oracle-ohasd.s
29
30  dg11      2022/11/19 17:02:04      /u01/app/grid/crsdata/dg11/olr/backup_20221119_17
31  2022/11/19 17:02:05 CLSRSC-327: Successfully configured Oracle Restart for a stand
32  [root@dg11 ~]#
33

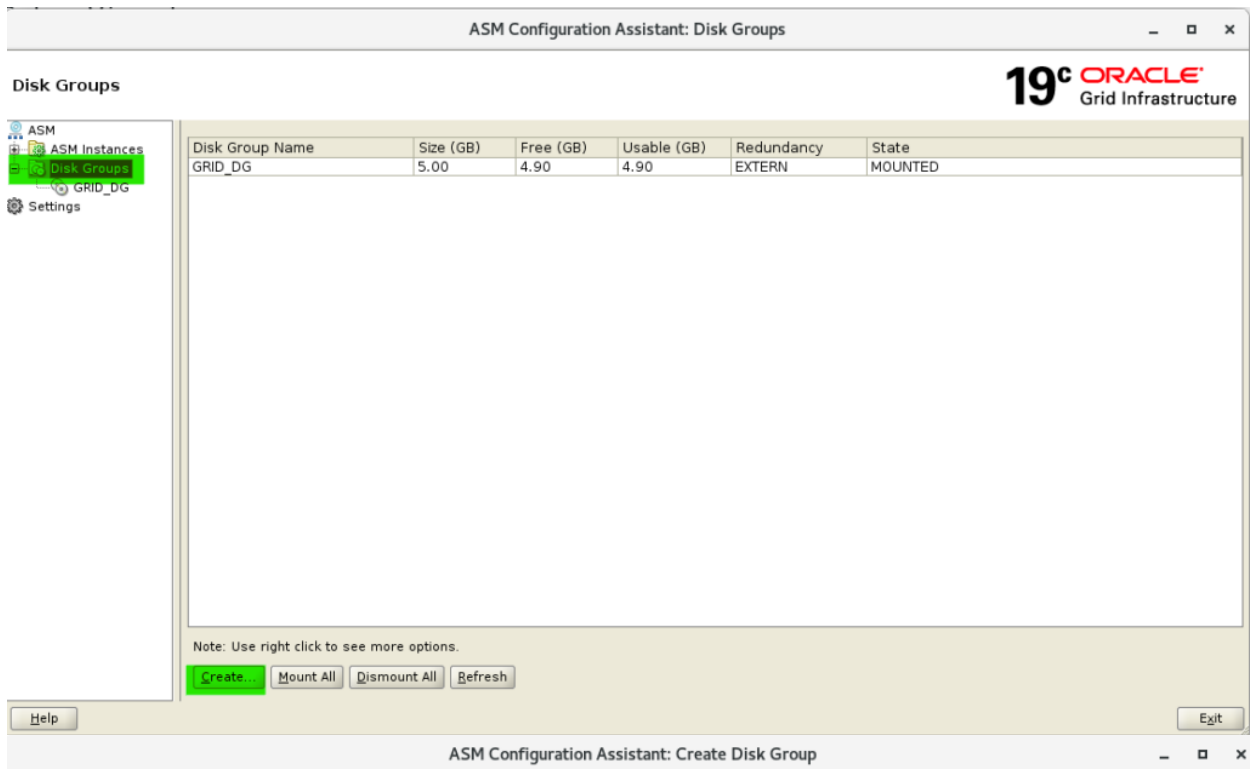
```

Then Click on OK for further progress of Grid installation.

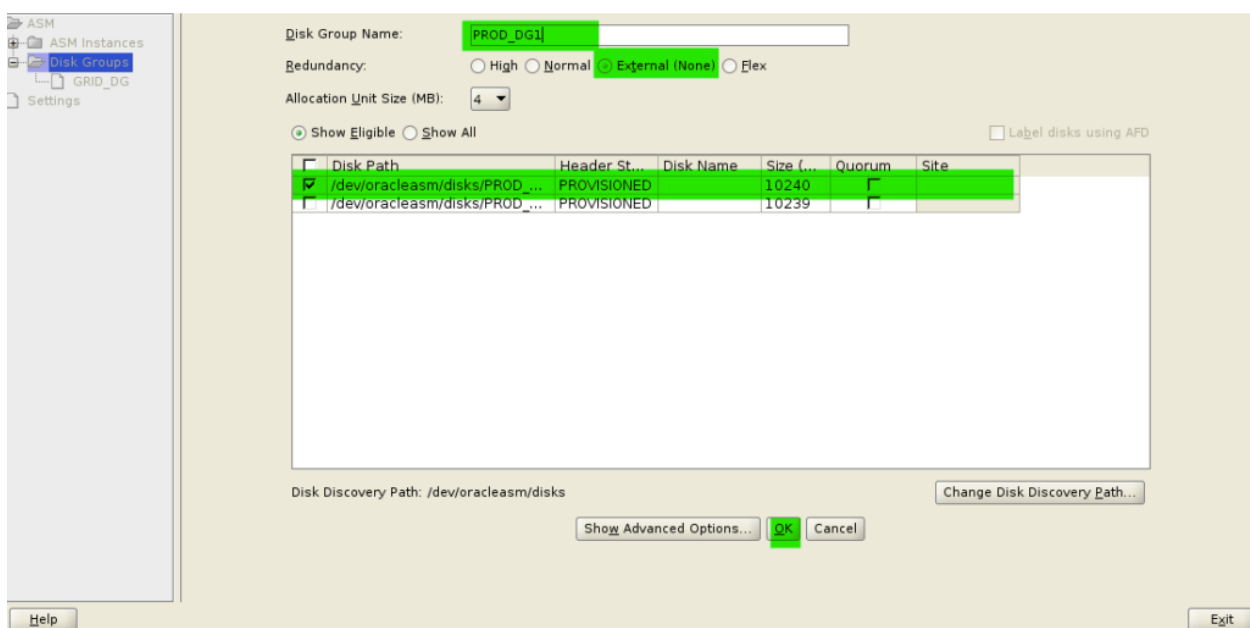


Configure ASM Disk groups

```
[grid@dgl1 bin]$ pwd
/u01/app/grid/product/19/grid/bin
[grid@dgl1 bin]$
[grid@dgl1 bin]$ ls -ltrh asmca
-rwxr-x---. 1 root oinstall 7.8K Nov 19 16:57 asmca
[grid@dgl1 bin]$
[grid@dgl1 bin]$ ./asmca
```



Create Disk Group



ASM Configuration Assistant: Disk Groups

Disk Groups

ASM

ASM Instances

Disk Groups

G...

PROD_DG1

Settings

Disk Group Name	Size (GB)	Free (GB)	Usable (GB)	Redundancy	State
GRID_DG	5.00	4.90	4.90	EXTERN	MOUNTED
PROD_DG1	10.00	9.91	9.91	EXTERN	MOUNTED

Note: Use right click to see more options.

CreateMount AllDismount AllRefresh

19cORACLE

Grid Infrastructure

ASM Configuration Assistant: Create Disk Group

Create Disk Group

ASM

ASM Instances

Disk Groups

G...

PROD_DG1

Settings

Disk Group Name: PROD_DG_FLASH

Redundancy: ☐ High ☐ Normal ☒ External (None) ☐ Flex

Allocation Unit Size (MB): 4

☒ Show Eligible ☐ Show All

☐ Label disks using AFD

☐ Disk Path	Header St...	Disk Name	Size (...)	Quorum	Site
☒ /dev/oracleasm/disks/PROD_...	PROVISIONED		10239	☐	

Disk Discovery Path: /dev/oracleasm/disks

Show Advanced Options...OKCancel

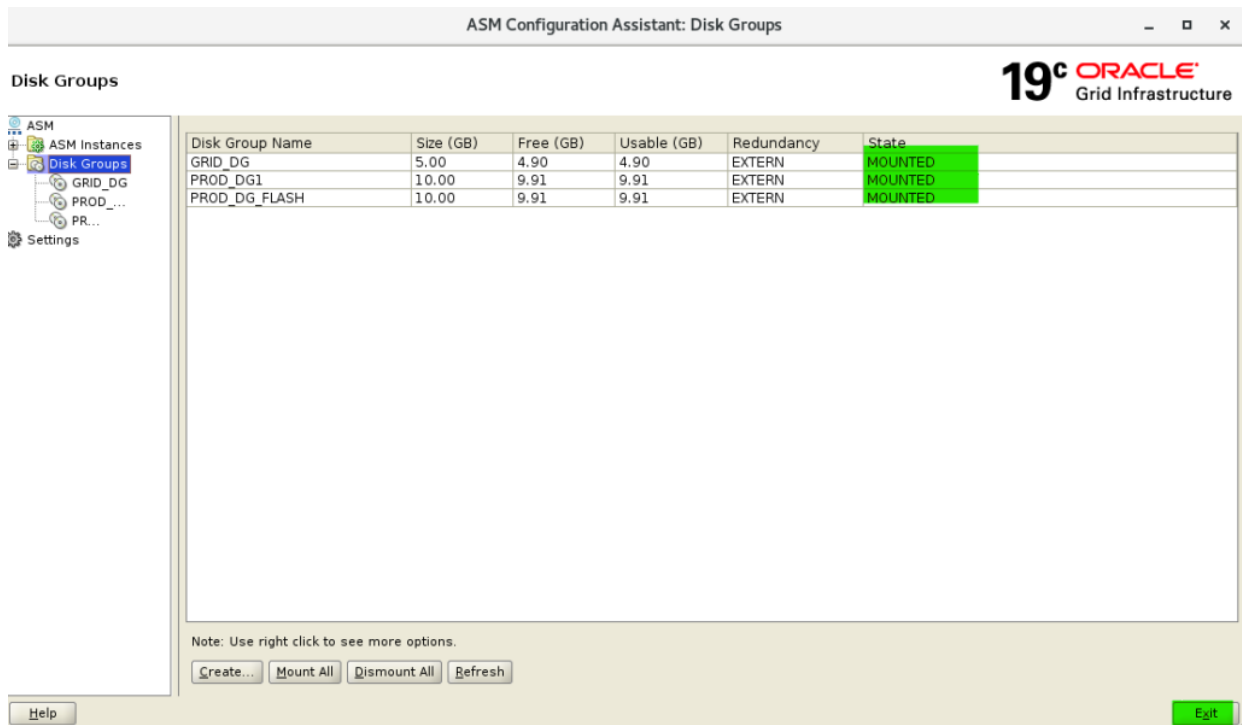
Change Disk Discovery Path...

Help

Exit

19cORACLE

Grid Infrastructure



```

1 [root@dg11 app]# su - grid
2 Last login: Sat Nov 19 17:02:04 IST 2022 on pts/1
3
4 [grid@dg11 ~]$ ps -ef|grep pmon
5 grid      14548      1  0 17:05 ?          00:00:00 asm_pmon_+ASM
6 grid      17046 16980  0 17:15 pts/1    00:00:00 grep --color=auto pmon
7 [grid@dg11 ~]$

```

Login to ASM & check number of Diskgroup Configured.

```

[grid@dg11 ~]$ . oraenv
ORACLE_SID = [+ASM] ? +ASM
The Oracle base remains unchanged with value /u01/app/grid
[grid@dg11 ~]$
[grid@dg11 ~]$ asmcmd
ASMCMD>
ASMCMD> lsdg
State      Type      Rebal    Sector  Logical_Sector  Block      AU      Total_MB  Free_MB  Req_mir_free_MB  Usable_file
MOUNTED    EXTERN    N        512     512             4096      4194304  5120     5020           0            1
MOUNTED    EXTERN    N        512     512             4096      4194304  10240    10152           0            1
MOUNTED    EXTERN    N        512     512             4096      4194304  10236    10148           0            1
ASMCMDS>

```